Experiment 1: Introduction to UNIX File System

Aim:

To study the structure of UNIX file system.

Theory:

UNIX file system is a hierarchical structure where all files and directories are organized in a tree-like structure starting from the root directory `/`.

Important Directories:

- `/` → Root directory
- `/home` → User files
- `/bin` → Essential commands
- `/etc` → Configuration files
- `/tmp` → Temporary files

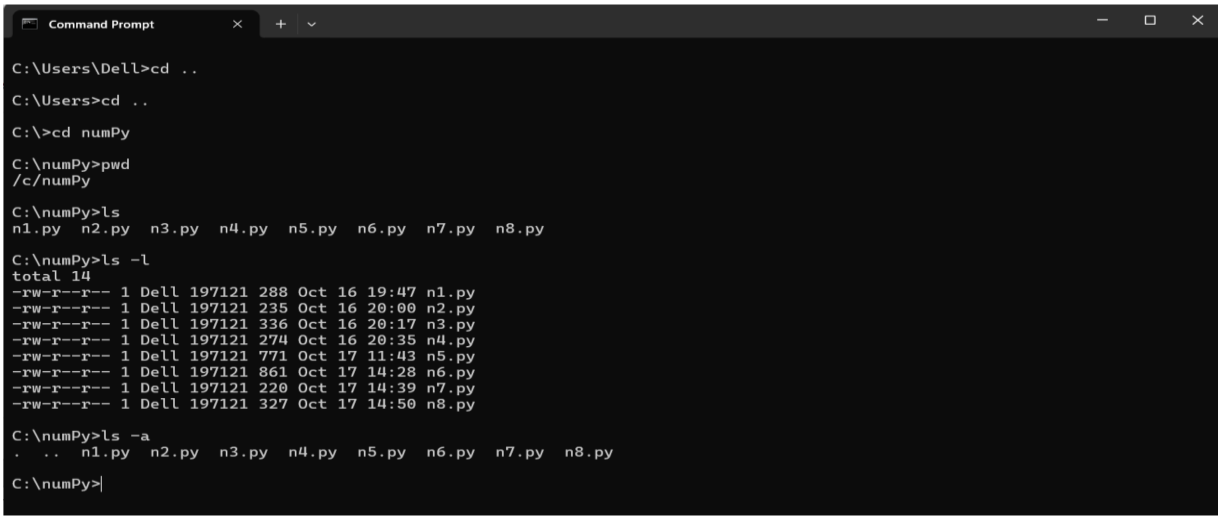
Commands:

```
pwd
ls
ls -l
ls -a
```

Procedure:

1. Open terminal.
2. Type `pwd` to check current directory.
3. Use `ls` to list files.
4. Use `ls -l` for detailed view.

Output:



```
Command Prompt

C:\Users\Dell>cd ..

C:\Users>cd ..

C:\>cd numPy

C:\numPy>pwd
/c/numPy

C:\numPy>ls
n1.py  n2.py  n3.py  n4.py  n5.py  n6.py  n7.py  n8.py

C:\numPy>ls -l
total 14
-rw-r--r-- 1 Dell 197121 288 Oct 16 19:47 n1.py
-rw-r--r-- 1 Dell 197121 235 Oct 16 20:00 n2.py
-rw-r--r-- 1 Dell 197121 336 Oct 16 20:17 n3.py
-rw-r--r-- 1 Dell 197121 274 Oct 16 20:35 n4.py
-rw-r--r-- 1 Dell 197121 771 Oct 17 11:43 n5.py
-rw-r--r-- 1 Dell 197121 861 Oct 17 14:28 n6.py
-rw-r--r-- 1 Dell 197121 220 Oct 17 14:39 n7.py
-rw-r--r-- 1 Dell 197121 327 Oct 17 14:50 n8.py

C:\numPy>ls -a
.  ..  n1.py  n2.py  n3.py  n4.py  n5.py  n6.py  n7.py  n8.py

C:\numPy>
```

Experiment 2: File and Directory Related Commands

Aim:

To learn file and directory operations.

Commands:

```
mkdir test
cd test
touch file1.txt
cp file1.txt file2.txt
mv file2.txt newfile.txt
rm newfile.txt
rmdir test
```

Procedure:

1. Create directory using `mkdir`.
2. Navigate using `cd`.
3. Create file using `touch`.
4. Copy, move, and delete files.

Output:

```
Command Prompt
C:\numPy>cd test

C:\numPy\test>touch file1.txt

C:\numPy\test>nano file1.txt

C:\numPy\test>cat file1.txt
echo "Hello World"

C:\numPy\test>cp file1.txt file2.txt

C:\numPy\test>cat file2.txt
echo "Hello World"

C:\numPy\test>mv file2.txt newfile.txt

C:\numPy\test>cat newfile.txt
echo "Hello World"

C:\numPy\test>rm newfile.txt

C:\numPy\test>rmdir test
The system cannot find the file specified.

C:\numPy\test>cd ..

C:\numPy>rmdir test
The directory is not empty.

C:\numPy>rm file1.txt
rm: cannot remove 'file1.txt': No such file or directory

C:\numPy>cd test

C:\numPy\test>rm file1.txt

C:\numPy\test>cd ..

C:\numPy>rmdir test
```

Experiment 3: Essential UNIX Commands

Aim:

To study basic UNIX commands.

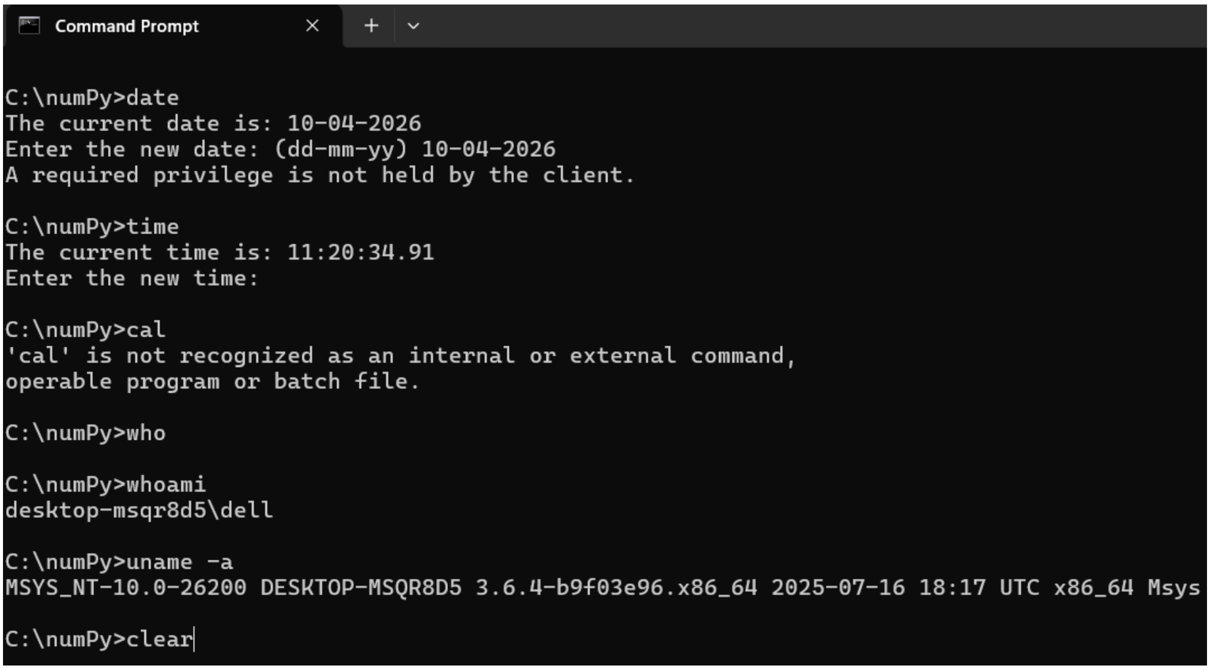
Commands:

```
date
cal
who
whoami
uname -a
clear
```

Procedure:

Execute each command in terminal and observe output.

Output:



```
Command Prompt

C:\numPy>date
The current date is: 10-04-2026
Enter the new date: (dd-mm-yy) 10-04-2026
A required privilege is not held by the client.

C:\numPy>time
The current time is: 11:20:34.91
Enter the new time:

C:\numPy>cal
'cal' is not recognized as an internal or external command,
operable program or batch file.

C:\numPy>who

C:\numPy>whoami
desktop-msqr8d5\dell

C:\numPy>uname -a
MSYS_NT-10.0-26200 DESKTOP-MSQR8D5 3.6.4-b9f03e96.x86_64 2025-07-16 18:17 UTC x86_64 Msys

C:\numPy>clear
```

Experiment 4: I/O Redirection and Piping

Aim:

To understand input/output redirection.

Commands:

```
echo "Hello" > file.txt
cat file.txt
cat file.txt >> file.txt
ls > list.txt
cat list.txt | sort
```

Explanation:

- `>` → overwrite
- `>>` → append
- `|` → pipe

Output:

Experiment 5: Introduction to VI Editor

Aim:

To learn VI editor basics.

Commands:

```
vi file.txt
```

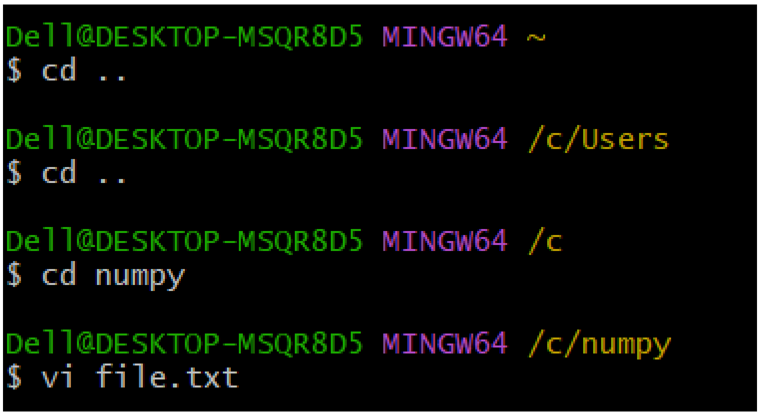
Modes:

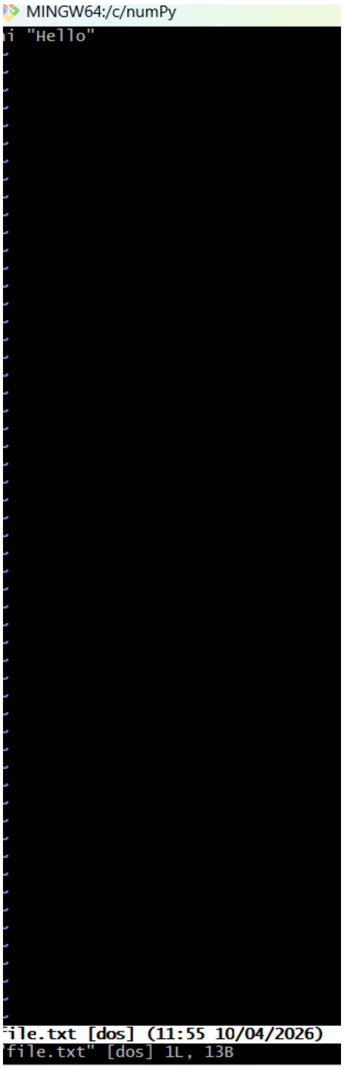
- Insert mode → press `i`
- Command mode → press `Esc`

Common Commands:

```
:w   (save)
:q   (quit)
:wq  (save & exit)
```

Output:





Experiment 6: Processes in UNIX

Aim:

To study process management.

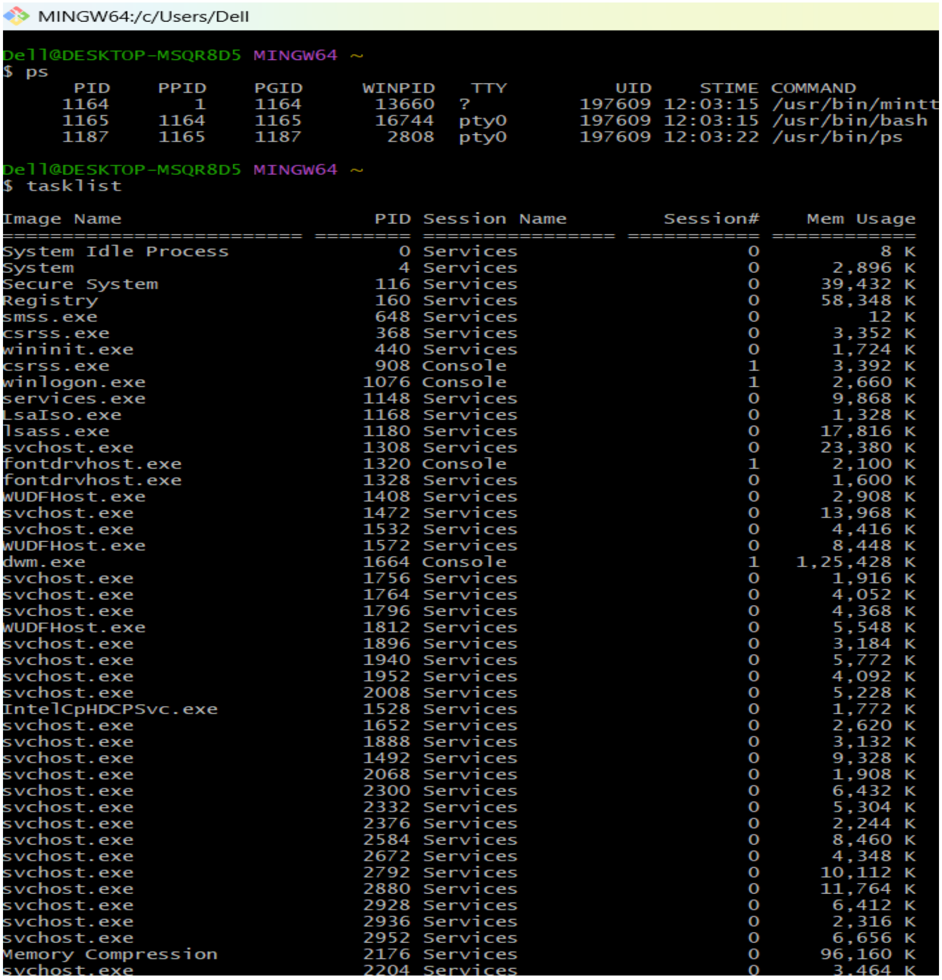
Commands:

```
ps
top
kill <PID>
```

Procedure:

1. Run `ps` to view processes.
2. Use `top` for live processes.
3. Kill a process using `kill`.

Output:



```
MINGW64:/c/Users/Dell

Dell@DESKTOP-MSQR8D5 MINGW64 ~
$ ps
      PID    PPID    PGID     WINPID   TTY         UID    STIME COMMAND
     1164       1    1164      13660  ?         197609 12:03:15 /usr/bin/mintt
     1165    1164    1165      16744  pty0      197609 12:03:15 /usr/bin/bash
     1187    1165    1187       2808  pty0      197609 12:03:22 /usr/bin/ps

Dell@DESKTOP-MSQR8D5 MINGW64 ~
$ tasklist

Image Name                     PID Session Name        Session#    Mem Usage
========================= ======== ================ =========== ============
System Idle Process              0 Services                   0          8 K
System                           4 Services                   0      2,896 K
Secure System                  116 Services                   0     39,432 K
Registry                       160 Services                   0     58,348 K
smss.exe                       648 Services                   0         12 K
csrss.exe                      368 Services                   0      3,352 K
wininit.exe                    440 Services                   0      1,724 K
csrss.exe                      908 Console                    1      3,392 K
winlogon.exe                  1076 Console                    1      2,660 K
services.exe                  1148 Services                   0      9,868 K
LsaIso.exe                    1168 Services                   0      1,328 K
lsass.exe                     1180 Services                   0     17,816 K
svchost.exe                   1308 Services                   0     23,380 K
fontdrvhost.exe               1320 Console                    1      2,100 K
fontdrvhost.exe               1328 Services                   0      1,600 K
WUDFHost.exe                  1408 Services                   0      2,908 K
svchost.exe                   1472 Services                   0     13,968 K
svchost.exe                   1532 Services                   0      4,416 K
WUDFHost.exe                  1572 Services                   0      8,448 K
dwm.exe                       1664 Console                    1   1,25,428 K
svchost.exe                   1756 Services                   0      1,916 K
svchost.exe                   1764 Services                   0      4,052 K
svchost.exe                   1796 Services                   0      4,368 K
WUDFHost.exe                  1812 Services                   0      5,548 K
svchost.exe                   1896 Services                   0      3,184 K
svchost.exe                   1940 Services                   0      5,772 K
svchost.exe                   1952 Services                   0      4,092 K
svchost.exe                   2008 Services                   0      5,228 K
IntelCpHDCPSvc.exe            1528 Services                   0      1,772 K
svchost.exe                   1652 Services                   0      2,620 K
svchost.exe                   1888 Services                   0      3,132 K
svchost.exe                   1492 Services                   0      9,328 K
svchost.exe                   2068 Services                   0      1,908 K
svchost.exe                   2300 Services                   0      6,432 K
svchost.exe                   2332 Services                   0      5,304 K
svchost.exe                   2376 Services                   0      2,244 K
svchost.exe                   2584 Services                   0      8,460 K
svchost.exe                   2672 Services                   0      4,348 K
svchost.exe                   2792 Services                   0     10,112 K
svchost.exe                   2880 Services                   0     11,764 K
svchost.exe                   2928 Services                   0      6,412 K
svchost.exe                   2936 Services                   0      2,316 K
svchost.exe                   2952 Services                   0      6,656 K
Memory Compression            2176 Services                   0     96,160 K
svchost.exe                   2204 Services                   0      3,464 K
```

```
Dell@DESKTOP-MSQR8D5 MINGW64 ~
$ kill 1187
```

Experiment 7: Communication in UNIX & AWK

Aim:

To study communication and AWK tool.

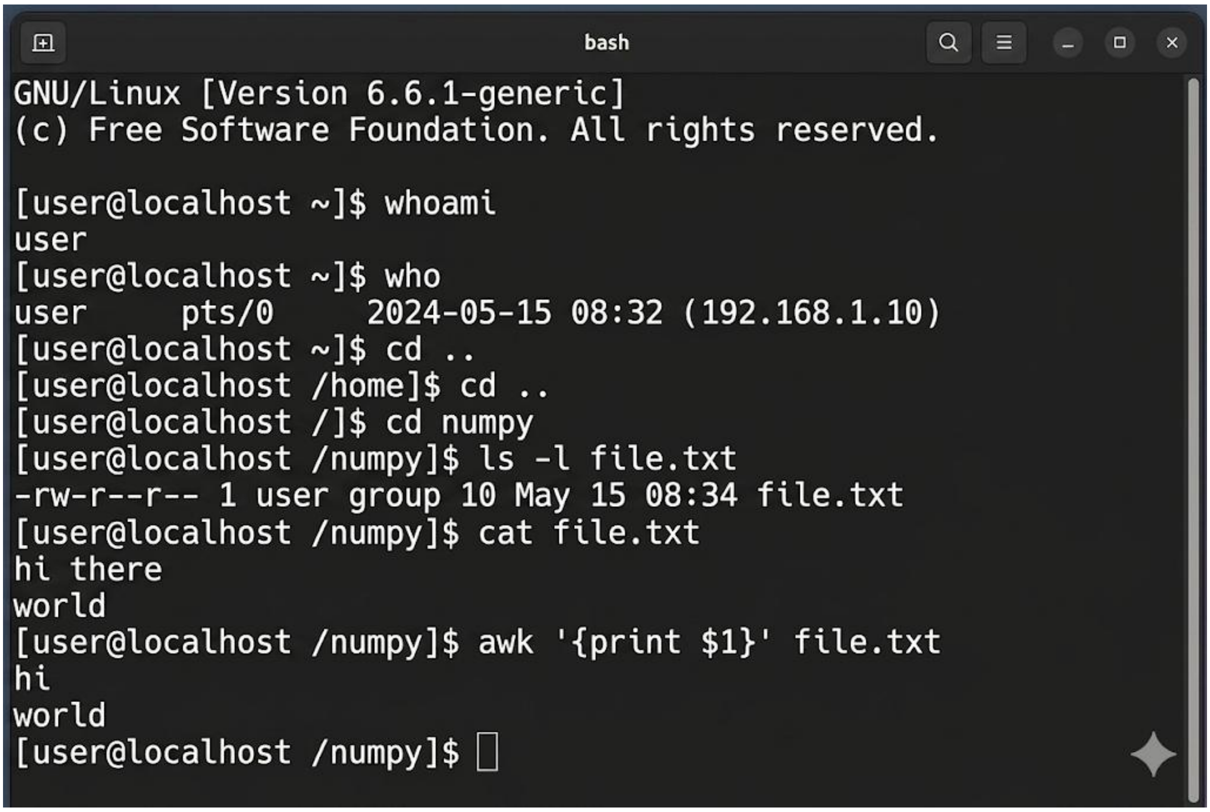
Commands:

```
who
write username
awk '{print $1}' file.txt
```

Explanation:

AWK is used for text processing.

Output:



Experiment 8: Shell Scripting Introduction

Aim:

To understand shell scripting.

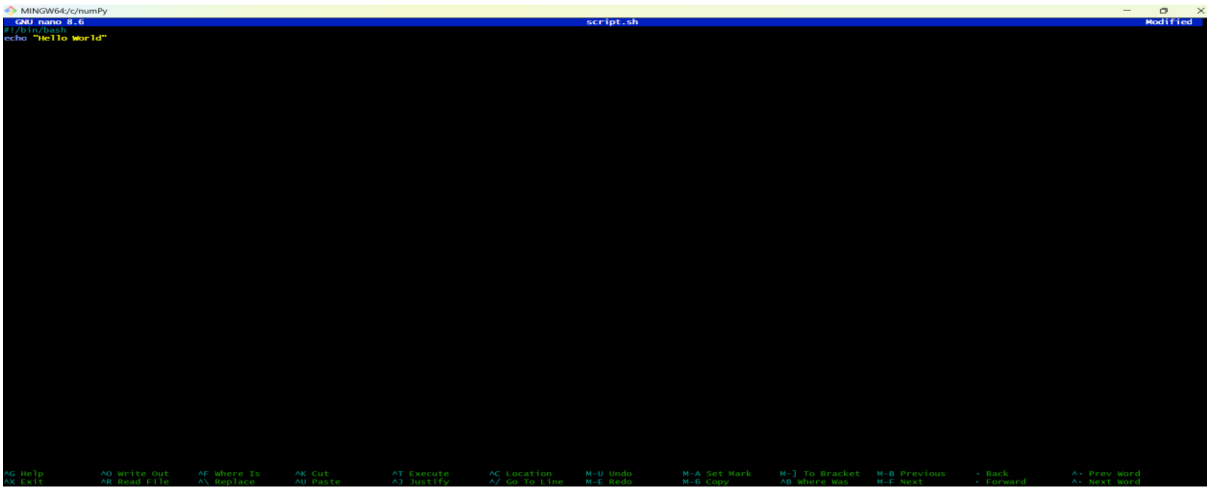
Example Script:

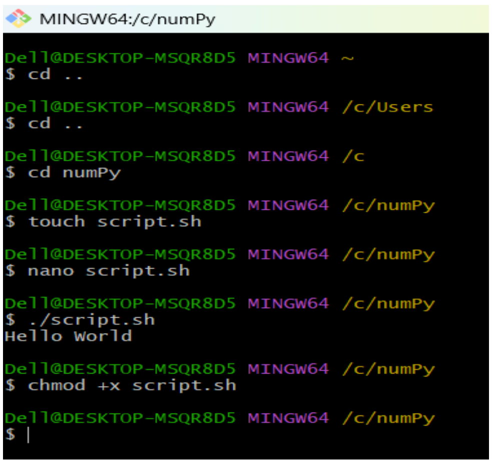
```
#!/bin/bash
echo "Hello World"
```

Run Command:

```
chmod +x script.sh
./script.sh
```

Output:





Experiment 9: Decision & Iteration in Shell

Aim:

To implement decision and loops.

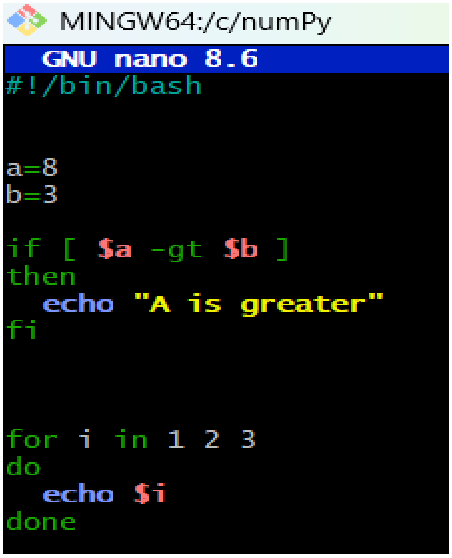
If Statement:

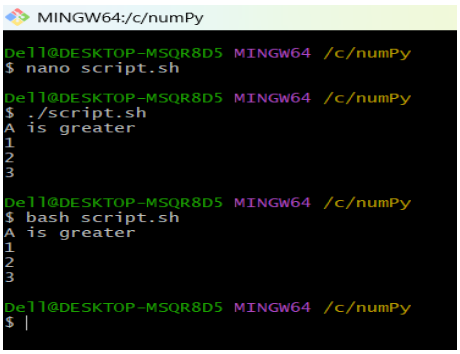
```
if [ $a -gt $b ]
then
echo "A is greater"
fi
```

Loop:

```
for i in 1 2 3
do
echo $i
done
```

Output:





Experiment 10: Shell Scripts for Problems

Aim:

To solve problems using scripts.

Example: Factorial Program

```
#!/bin/bash
n=5
fact=1
for((i=1;i<=n;i++))
do
fact=$((fact*i))
done
echo $fact
```

Output: